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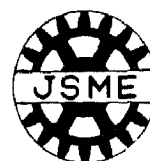
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IN BUNDLE VOID FRACTION MEASUREMENT OF PWR FUEL ASSEMBLY

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ABSTRACT

Void fraction measurement test for PWR fuel assembly has been being carried out from 1987 as a Japanese national project. This test is one of programs of the various proof/verification tests on safety and reliability of nuclear power plants. In this test, the void fraction is measured by the gamma-ray transmission method.

The preliminary test on the void measuring system, the design and construction of the facility and the test operation of the whole facility have been completed and the function of loop control system and that of void fraction measuring system were confirmed to be sufficient to start this test program. This test is now in progress and will continue until 1994 based on the void measuring method described in this paper.

INTRODUCTION

The Nuclear Power Engineering Corporation (NUPEC) is performing the various proof/verification tests on safety and reliability of nuclear power plants under the sponsorship of the Ministry of International Trade and Industry (MITI) as a national project. As one of programs in this project, the void fraction measurement test of a pressurized water reactor (PWR) fuel assembly has been being carried out from 1987.

The preliminary test on the void measuring system, the design and construction of the facility and the test operation of the whole facility have been completed by Mitsubishi Heavy Industries, Ltd.. The results of the preliminary test are reflected in the design. The test program is now in progress and will continue until 1994. This paper describes the test plan, the test facility and the typical results of preliminary test and test operation.

The core behavior to void generation is shown in Figure 1. The void fraction in PWR core is negligible under the normal condition, but, becomes remarkable in the hot assembly under the anticipated transient or the accident condition. The void generation

introduces the redistribution of flow through the fuel assemblies, while it reduces the core peaking factor due to the void reactivity feed-back. As such, the void behavior affects the thermal and nuclear characteristics of the reactor core and is one of the most important factors in the core design. Therefore,

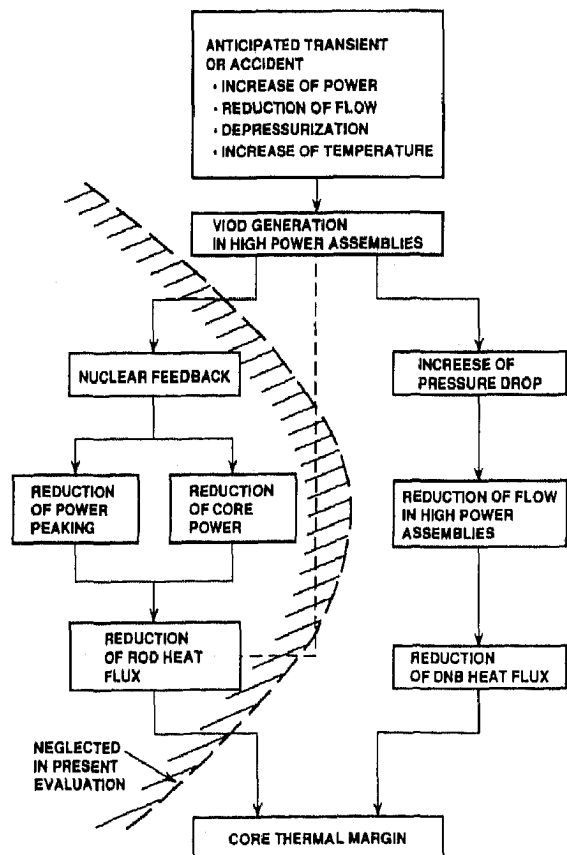


Figure 1 Core Behavior to Void Generation

even in PWR, it is important to understand the void fraction for the evaluation of core thermal-hydraulics.

The void reactivity feed-back and its effects shown on the left of a curved line in figure 1 are not taken into consideration in the present PWR core design. The purposes of this test are as follows.

(1) Verification of Void Correlation and Improvement of Subchannel Analysis Code

(2) Development of Advanced Core Design Methodology Including Reactivity Feed-back by Void Generation

This test contains the rod bundle experiments and the subchannel experiments. The void fraction in each experiment is measured by the gamma-ray transmission method.

The subchannel experiments are carried out to investigate the void distribution effect on the subchannel averaged void fraction measurement. In these experiments, the void fraction in the subchannel simulating one subchannel of rod-bundle is measured by the wide gamma-ray beam densitometers and the narrow gamma-ray beam computed tomography (CT) scanner at the heated section. Both results give the relationship between the chordal averaged void fraction and the subchannel averaged void fraction. This relationship is used to modify the measured results of rod bundle experiments to obtain the representative void fraction of each subchannel.

The bundle experiments are carried out to obtain the void-fraction of each subchannel. Test bundles are fabricated simulating partial section and full length of the 17x17 type PWR fuel assembly. Electrically heated skin heaters are used as the models of fuel rods. In these experiments, the void fractions of 36 subchannels are obtained by the gamma-ray multi-beam system at three different elevations of test bundle.

The test facility is designed to simulate the thermal-hydraulic conditions in the actual PWR core including the steady state and the transients.

TEST PROCEDURE

The transmission method of gamma-ray is adopted in this study to measure the density/void-fraction of the gas-liquid two-phase flow.

Figure 2 shows the test procedure. The subchannel experiments and the rod bundle experiments are performed in this test. The test section for the subchannel experiments is simulating one of three type subchannels of the rod bundle, a center subchannel, a side subchannel and a corner subchannel. The narrow gamma-ray beam CT scanner gives the subchannel averaged void fraction and the transmission information of the wide gamma-ray beam gives the chordal averaged void fraction. The subchannel test gives the relationship between these two averaged values. Such relationship is obtained individually for each type subchannel.

On the other hand, the CT scanner can not be applied to the rod bundle test from a view-point of the ability of gamma-ray transmission. So, each subchannel averaged void fraction of the rod bundle is not given only by the wide gamma-ray multi-beam system. Therefore, the subchannel averaged void fraction of the rod bundle is introduced by both the results obtained by the rod bundle experiments and the above relationship given by the subchannel experiments.

TEST FACILITY

The void fraction measurement test is being

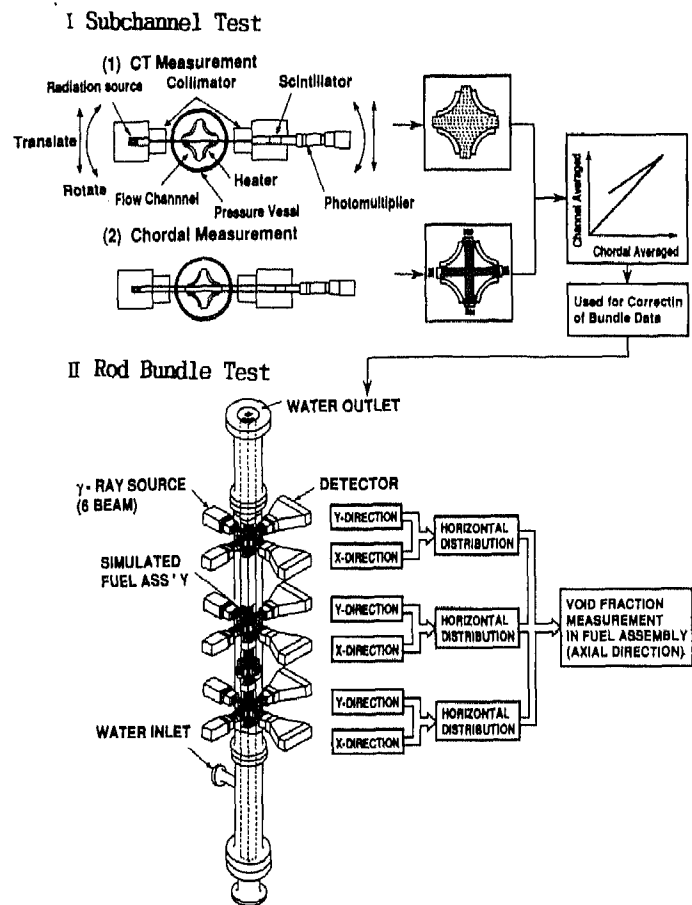


Figure 2 Test Procedure

performed at the high pressure and high temperature loop constructed in Takasago Engineering Laboratory of NUPEC, which can simulate the PWR thermal-hydraulic conditions including the steady state and the transients as the power increase, the flow reduction, the depressurization and the temperature increase. The loop used for the critical heat flux (CHF) test of Japanese PWR fuel assembly (Makihara et al., 1990) was partially reconstructed for this test. The CHF test section was removed and a subchannel test section, a rod bundle test section and a void measuring system have been added for this test.

The system diagram of this loop is shown in Figure 3. This loop consists mainly of two systems, a circulation system and a cooling system. The circulation system supplies the high temperature pressurized water in fixed temperature, pressure and flow rate for each test section. The cooling system removes the heat added to the water in the loop and control the temperature and the pressure at any given conditions. This loop has two test sections used for the subchannel test or the rod bundle test.

In the case of the subchannel test, the rod bundle test section is used as the bypass line of the flow fed by the circulation pump having the big capacity relatively to the subchannel test section. On the other hand, the subchannel test section is isolated from the circulation system in the rod bundle test.

The flow rate through each test section is controlled by a oil pressure driven control valve and measured by a turbine flow meter in the flow control section upstream of each test section. The water temperature is controlled by a preheater and measured by a calibrated platinum resistive thermometer (RTD) at the inlet of test section. The pressure is

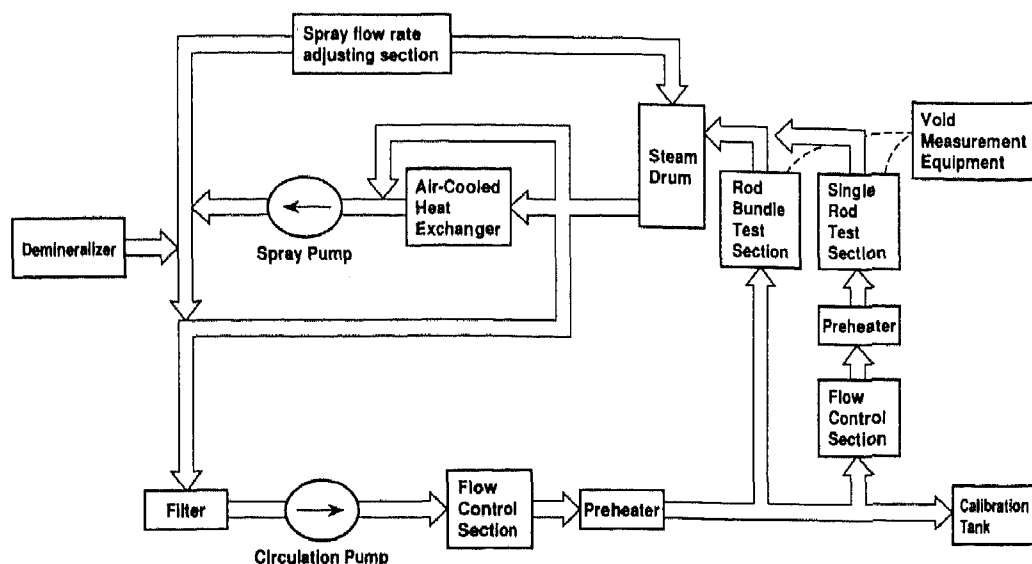


Figure 3 System Diagram of Test Facility

controlled by the adjustment of heater output and subcooled water spray flow rate of the steam drum. The test section inlet pressure is measured by a high precision quartz manometer. These controllings are performed by a direct digital control system.

The degassed pure water is supplied to the loop as the coolant and the main equipments are made of stainless steel to prevent the water from fouling.

The design pressure is 19.2 MPa and the design temperature is 362°C. The parameter ranges covers:

Pressure:	4.9 - 16.6 MPa
Inlet Temperature:	140 - 345°C
Mass Velocity:	550 - 4150 kg/m ² s
The transient parameters are as follows.	
Depressurization:	-0.03 MPa/s
Temperature Increase:	1 °C/s
Flow Reduction:	-25 %/s
Power Increase:	15 %/s
The accuracy of the test parameters is estimated as:	
Pressure:	1 %
Inlet Temperature:	1 °C
Flow:	1.5 %
Test Section Power:	1 %

TEST SECTION

Figure 4 shows the subchannel test section and Figure 5 shows the rod bundle test section. The test section consists of a pressure vessel, a flow channel and a simulated subchannel/fuel-assembly. The electrically heated skin heaters are used as the models of fuel rods.

A subchannel model simulating a center subchannel, a side subchannel or a corner subchannel of the test rod bundle is installed in the subchannel test section. The axially cut tube of Inconel alloy 600 is used as a heater element. The axial power distribution is uniform. The effective heated length is 1.5 m where the void measuring section is set near this top end. The gamma-ray CT scanner system or the gamma-ray chordal densitometer system is attached around this section.

Test bundles used in the rod bundle test are fabricated simulating partial section and full length

of the 17x17 type Japanese PWR fuel assembly. These are arranged in 5x5 square array. The test bundle consists of Inconel alloy 600 tubes as the electrically heated skin heaters, the grid spacers and the electrodes. The axial power distribution is uniform or cosine profile. The void measuring sections are set at three different elevations of test bundle.

The void measuring section of the pressure vessel is made of titanium alloy and the wall thickness where the gamma-ray beam transmits is cut down as in Figure 4 and 5 to minimize the gamma-ray attenuation in the structure.

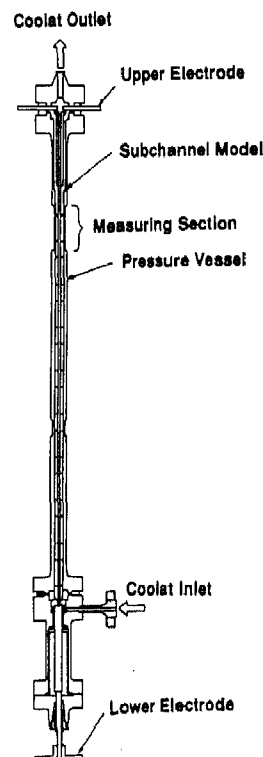


Figure 4 Test Section for Subchannel Test

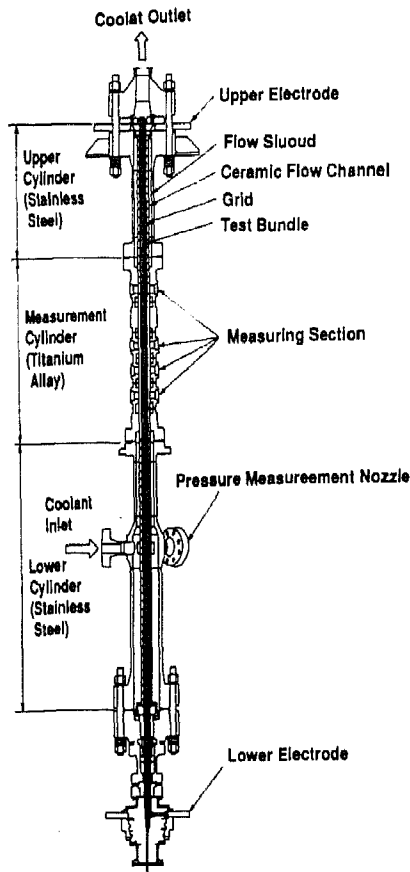


Figure 5 Test Section for Rod Bundle Test

VOID MEASURING

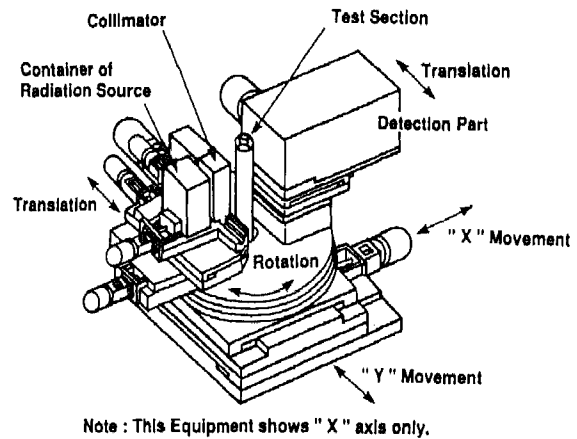
The void measuring systems are shown in Figure 6 and 7. These systems consist mainly of the gamma-ray sources, the detection parts, the collimators and the signal processing unit. The radio isotope ^{137}Cs is applied as the stable gamma-ray source and the gamma-ray is collimated to chordal beam. The attenuation of the gamma-rays dependent on the void fraction can be detected with this system. The detection part consists of a collimator, a scintillator, a light guide, a photomultiplier and a preamplifier. The output signal of preamplifier is fed to the signal processing unit working in the pulse counting mode in the control room. The crystal of cerium-doped gadolinium silicate (GSO) is used as the scintillator which has very short decay time of scintillation light suitable for the high count rate measuring. The detection part is put in the constant temperature circumference to minimize the drift of the output due to the temperature fluctuation.

The intensity of the gamma-ray source was determined to get the measurable count-rate of the signal processing, that is $30 \times 10^4 \text{ cps}$.

Three void measuring systems are prepared as follows.

(1) Void fraction measuring systems for subchannel test (Figure 2 and 6)

(a) Subchannel averaged void fraction: The CT scanner system gives the distribution of density/void-fraction over the subchannel under the steady state condition. This distribution data introduce the subchannel averaged void fraction. Two gamma-ray beams are used to reduce the measuring time. The CT system is operated by the



Note : This Equipment shows "X" axis only.

Figure 6 Void Measuring System for Subchannel

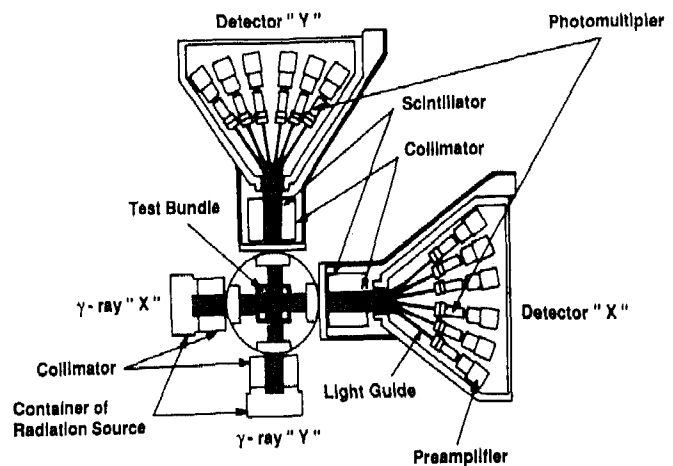


Figure 7 Void Measuring System for Rod Bundle

translate/rotate method as shown in Figure 6. At each translation/rotation location, the intensities of gamma-rays attenuated by the object called "projection data" are detected by the detector. The projection data needed for the reconstruction of two-dimensional properties are obtained by the translation/rotation around the object. The image reconstruction is performed by the filtered back-projection algorithm to give the distribution of the linear attenuation coefficient. The spatial resolution of this CT is approximately 0.5 mm. In order to avoid the effect of the motion of flow and obtain the time averaged void fraction, the sufficient measuring time is needed. This needed measuring time is confirmed by checking the dependency of the subchannel averaged void fraction obtained by the actual measurement on it.

(b) Chordal averaged void fraction: Two densitometer systems give the x-direction chordal averaged void fraction and that of y-direction as shown in Figure 6. Each systems consist of the gamma-source and the detection part. These two systems are located on the same tables as CT. The translation/rotation tables are fixed in this case. These systems can be applied for both the steady state test and the transient test.

(2) Void fraction measuring system for rod bundle test (Figure 2 and 7): The multi-beam system are used to obtain each subchannel void fraction of the rod bundle. The six transmission data of

x-direction and those of y-direction between rod and rod/channel wall are used to reconstruct the void fractions of 36 subchannels by the iterative method. These subchannel void fractions correspond to those of the chordal measurement of the subchannel test. The relationship between the subchannel averaged void fraction and the chordal averaged void fraction obtained by the subchannel test is used to introduce the subchannel averaged void fraction of the rod bundle as mentioned in TEST PROCEDURE.

Such measurements are performed simultaneously at three elevations as shown in Figure 2 and the axial distribution of void fraction is obtained.

PRELIMINARY TEST

The preliminary test on the void measuring system was performed and the results have been reflected in its design. This test contains the next items.

(1) The effect of the circumferential temperature fluctuation on the stability of detector output was checked to determine the required specifications of the circumferential temperature control and to select the measuring mode.

(2) The effect of the magnetic field fluctuation on the stability of the detector output was checked to determine the specifications of the magnetic shield. The magnetic field is made by the large electrical current fed to the skin heaters of the test section.

(3) The gamma-ray scattering effect was checked to determine the collimator specifications and to select the measuring mode.

The obtained results were used to determine the system specifications. Main reflections are as follows.

(1) The circumferential temperature of detector is controlled to be at the set value $\pm 0.3^\circ\text{C}$.

(2) The pulse counting mode is adopted which can mitigate well the effect of the scattered gamma-rays and the circumferential temperature fluctuation as compared with the current mode.

(3) The collimator is designed to minimize the scattering effect. The tungsten collimator of 100 mm in depth is used.

(4) The photomultiplier and the preamplifier are covered by the multi-layered shield to prevent the effect of the magnetic field fluctuation.

Further, the accuracy was confirmed by the measuring the mock-up simulating the structure of test section and the acrylic resin phantom as the model of two-phase flow. Figure 8 shows the dimensions of the subchannel and the examples of the acrylic resin phantoms simulating the bubble flow and the plug flow in the center subchannel for the subchannel test. Figure 9 shows all phantoms used in the preliminary test. The results shows that the

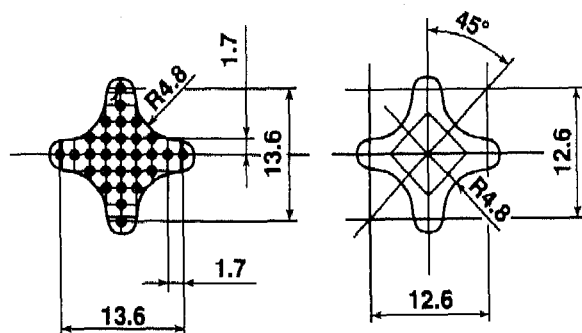


Figure 8 Subchannel Model

accuracy of void fraction is less than the required accuracy which is about 5% as shown in Figure 10. It is also expected from this figure that the correction of the chordal averaged void fraction to the subchannel averaged void fraction is less than about 5% in void fraction.

TEST OPERATION

The test operation was performed in 1991-1992 after the completion of test facility construction to

Void Fraction	Phantoms	Measurement
0 %	Acrylic Resin Simulating Water	—
18.9 %	Void	CT Chordal
28.4 %		CT Chordal
26.0 %		Chordal
24.2 %	Void	CT Chordal
26.1 %		Chordal
44.3 %	Void	CT Chordal
55.4 %		CT Chordal

Figure 9 Simulated Two-phase Flow in Subchannel

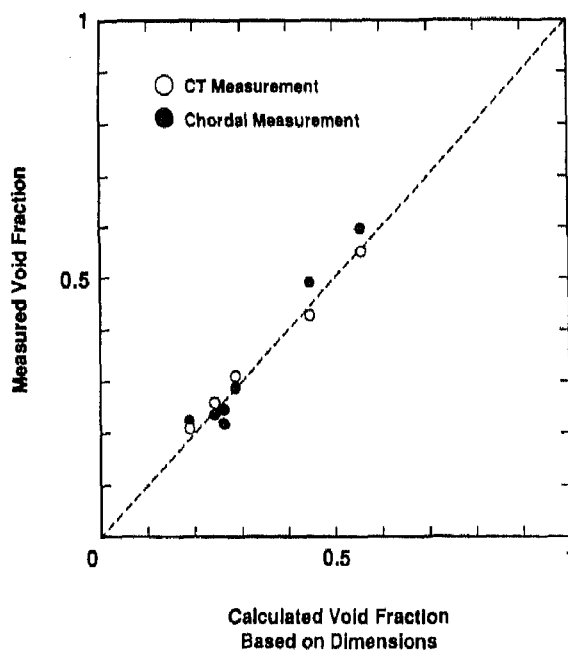


Figure 10 Verification Test Results

confirm the functions of the high temperature and high pressure loop and the void fraction measuring systems.

The loop was operated under the high temperature and high pressure conditions and confirmed to be controlled in the fixed steady or transient test conditions by the loop controlling system. Three measuring systems were tried to measure the water single-phase flow and the steam-water two-phase flow under the high temperature and high pressure conditions. And, it was confirmed that the density of single phase flow and the void fraction of two-phase flow can be obtained by the data processing system as designed.

Several results of test operation are described as followings.

The typical results of the local void fraction distribution of the subchannel test by the CT scanner system are shown in Figure 11. These are measured at the measuring section near the end of the heated section shown in Figure 4 where is 1.4m from the beginning of the heated section. In the low quality case of 1.6% under 14.7 MPa in pressure as Figure 11(a), the flow pattern is expected to be bubbly flow and the steam bubbles gather near the heated wall. On the other hand, the annular-mist flow is expected in the relatively high quality case of 19% under 9.90 MPa in pressure as Figure 11(b).

Figure 12 shows the results of the experiment under the depressurization transient condition in the subchannel test. The chordal averaged void fraction increases as the depressurization which is approximately -0.03MPa/s .

Figure 13 and 14 show the results of the experiment under the flow reduction transient condition in the rod bundle test. The flow transient and the other parameters such as the temperature, the pressure and the test section power controlled at the fixed values are shown in Figure 13. The distribution of the subchannel void fraction and the bundle averaged void fraction are shown in Figure 14 with the time interval 0.5 second. The test bundle has the uniform axial power distribution. The measuring section is the top one in Figure 5. This result is trial one of the data processings mentioned in the section "VOID MEASURING" without the correction of relationship between the subchannel averaged void fraction and the chordal averaged void fraction. It is found that the void fraction increases with the flow reduction. The bundle averaged void fraction is approximately 0% at the beginning of the flow reduction transient and becomes 44% at the end of the transient.

The comparison of the density of water single phase flow under the high pressure and high temperature conditions obtained by three measuring systems with the steam table showed that the maximum relative error is less than 2 %. Figure 15 shows the measured results on the distribution of subchannel density of the water single phase flow of 14.7 MPa in pressure and 300 C in temperature. The density of each subchannel is almost uniform and to be nearly equal to the value 0.725g/cm^3 given by the steam table.

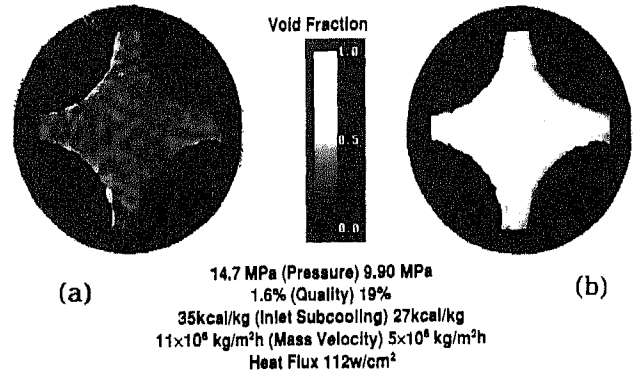


Figure 11 Void Distribution in Subchannel by CT

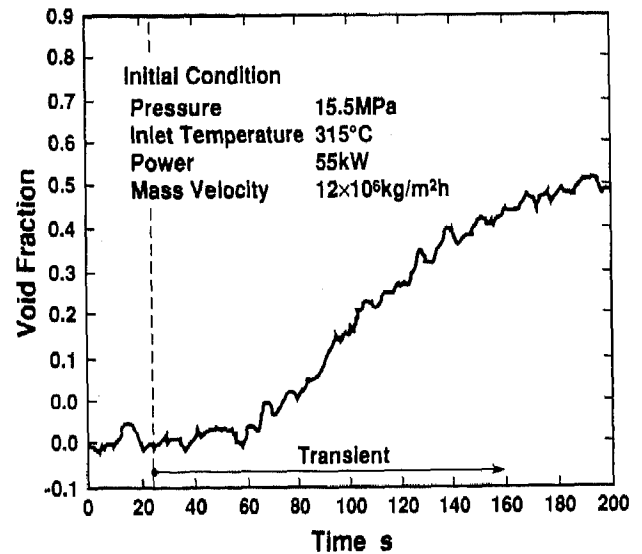
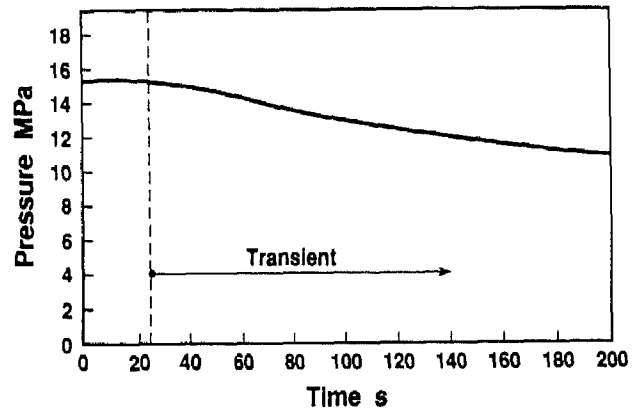


Figure 12 Depressurization Transient

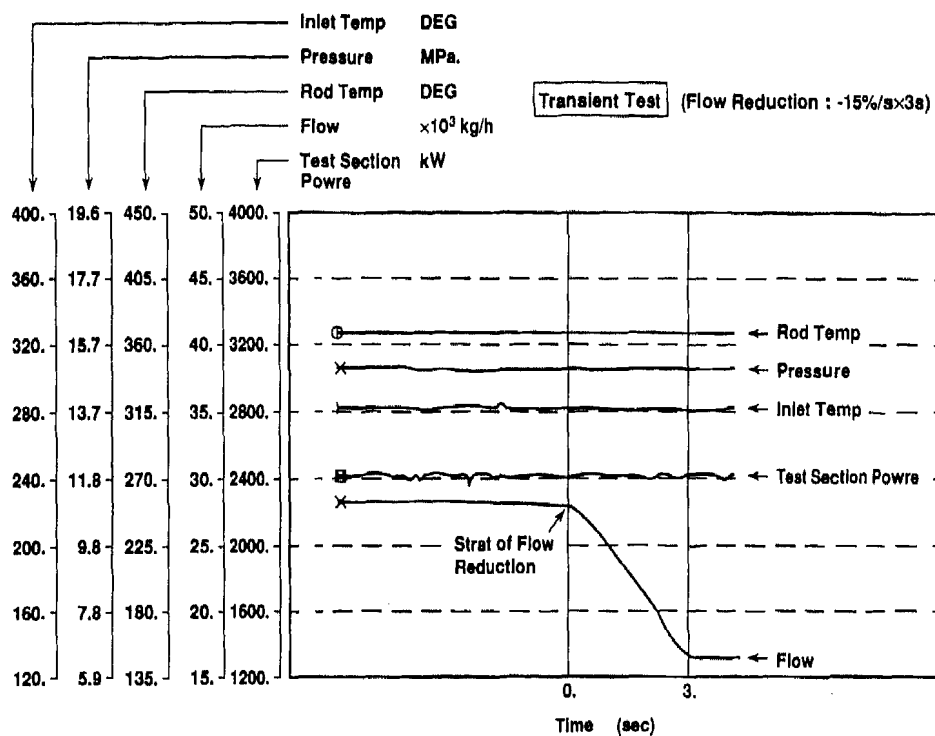


Figure 13 Flow Transient in Rod Bundle Test

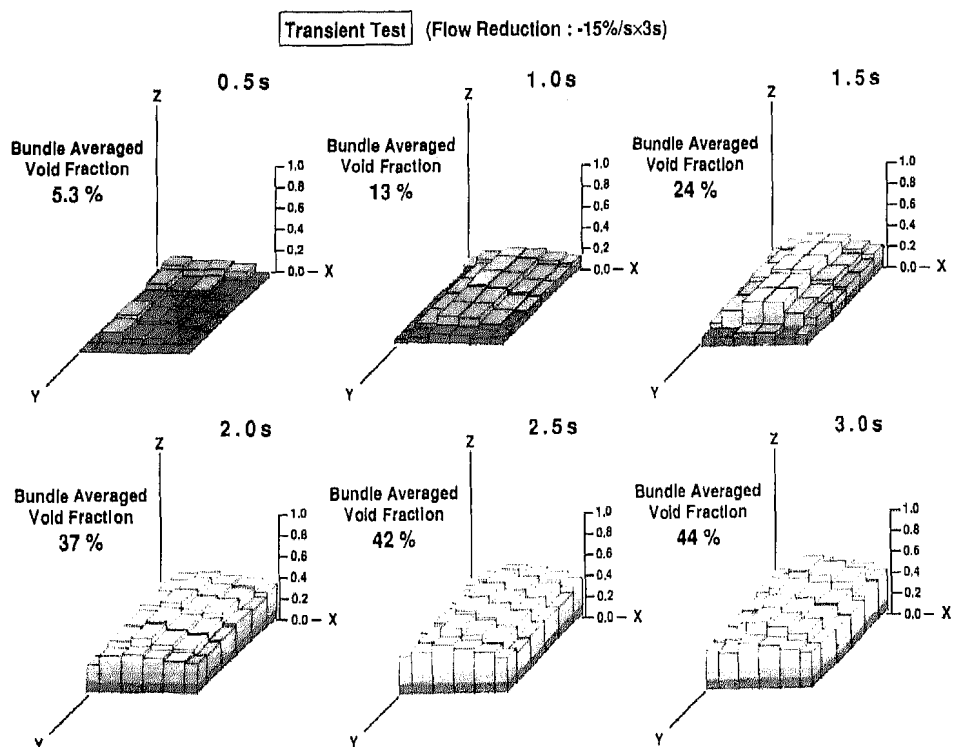


Figure 14 Void Fraction Distribution in Flow Transient

PROOF TEST

This proof test program is now in progress and will continue until 1994. The test items are as shown in Table 1. The steady and transient subchannel/rod-bundle experiments are planned and the results of subchannel test will be reflected in the data processing of rod bundle test. The results on the void fraction of rod bundle will be used to evaluate the void correlations and the subchannel analysis method for PWR core thermal-hydraulic design.

CONCLUSION

The function of the void fraction measuring systems and that of the loop control systems are confirmed to be sufficient to start this proof test by the preliminary test and the test operation. The following major conclusions are obtained.

(1) The accuracy was confirmed to be sufficient for this proof test.

(2) The subchannel averaged void fraction of rod bundle simulating the PWR fuel assembly can be obtained by this gamma-ray multi-beam system under the correction rule introduced by the subchannel test.

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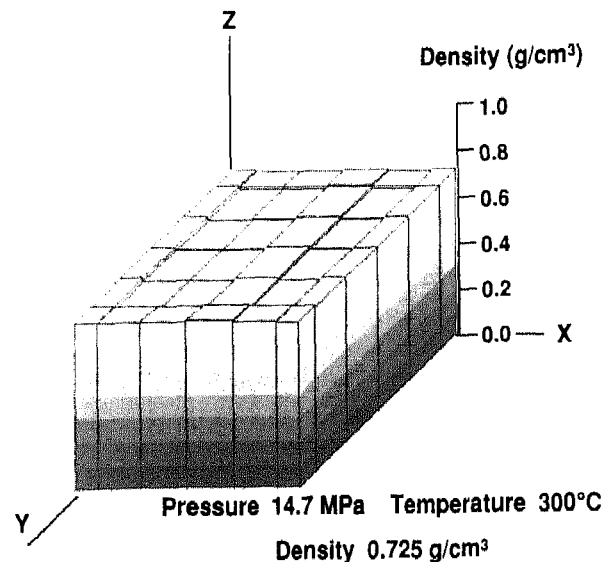


Figure 15 Density Distribution

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Table 1 Test Items of Proof Test

Test Section	Type of Cell	Axial Power Distribution	Test Mode
Subchannel	Center Channel (Typical)	Uniform	Steady State / Transient
	Center Channel (Thimble)	Uniform	Steady State / Transient
	Side Channel	Uniform	Steady State
	Corner Channel	Uniform	Steady State
Rod Bundle	Typical Cell	Uniform	Steady State / Transient
	Typical Cell	Cosine	Steady State / Transient
	Thimble Cell	Cosine	Steady State / Transient